

Cross-Method Saliency Framework: Motivation, Notation, and Findings

Why cross-compare at all?

No single token-attribution method gives ground-truth importance. Each method is a different, imperfect lens on what a model is doing:

- **Attention models** (LSTM, BiLSTM, 2×BiLSTM with attention): we extract the argmax tokens from the attention layer. This tells us where the attention head concentrates weight, but attention weights are not reliably causal—a token can receive high attention without being decision-relevant, and the relationship between attention and prediction is contested in the literature.
- **TF-IDF classifiers** (Naïve Bayes, Random Forest, Logistic Regression): these operate on bag-of-words TF-IDF vectors, so they are blind to word order, context, and sequential patterns. Importance is read from model coefficients or feature rankings, which reflect corpus-level statistical association, not message-level reasoning.
- **Character n -gram TF-IDF Logistic Regression**: operates on sub-word character sequences, capturing morphological and orthographic patterns (e.g. obfuscated spam tokens like “fr33” or “w1n”) but is even further removed from semantic content.

Because each method is limited in a *different* way, the interesting question is not “which tokens does model X think are important?” but rather: **when we look across mechanistically different models, where do they agree and where do they disagree?** Agreement across flawed-but-differently-flawed methods is more credible than any single method’s output. Disagreement tells us something is being captured by one attribution family that the other is structurally blind to.

Universe and attribution sets

Let

$$U = \{\text{all tokens in a given message}\}$$

be the universe of discourse for one message. For each model, salient tokens are extracted by a top- k selection process applied to the model’s token-importance scores (attention weights for the attention family; TF-IDF feature importance, coefficients, or rankings for the classical family).

Let

$$A \subseteq U$$

denote the union of top- k tokens selected across the **attention-based** models. Here A stands for *attention*.

Let

$$R \subseteq U$$

denote the union of top- k tokens selected across the **robust classical comparison pool**. Here R stands for *robust*.

Both A and R are constructed as unions across their respective model families: a token enters A if *any* attention model selects it in its top- k , and likewise for R . This is a deliberately permissive construction—it asks “did at least one model in this family consider this token salient?” rather than requiring unanimous agreement within a family.

Core decomposition

The message-level salience structure is decomposed into three pairwise-disjoint regions whose union equals $A \cup R$.

1. Shared set

$$S = A \cap R$$

Tokens selected by *both* attribution families. These are the cases of cross-method convergence—where attention and classical methods independently point to the same token.

2. Attention-exclusive set

$$E_A = A \setminus R = \{x \in U \mid x \in A \text{ and } x \notin R\}$$

Tokens selected by attention **but not** by the classical pool. These are tokens that attention prioritises but that do not surface through TF-IDF-based importance—potentially because they carry sequential, positional, or contextual signal that bag-of-words representations cannot capture.

3. Robust-exclusive set

$$E_R = R \setminus A = \{x \in U \mid x \in R \text{ and } x \notin A\}$$

Tokens selected by the classical pool **but not** by attention. These are tokens with strong corpus-level statistical association to a class that attention does not concentrate on—potentially because the attention mechanism has learned to distribute weight elsewhere or because the signal is redundant given other tokens in the sequence.

Partition property

The three regions S , E_A , E_R are pairwise disjoint and exhaustive over the attributed region:

$$S \cup E_A \cup E_R = A \cup R, \quad S \cap E_A = S \cap E_R = E_A \cap E_R = \emptyset.$$

What the decomposition reveals

The point of the framework is not the notation—it is what falls out when you apply it to real messages.

- S (shared tokens) tends to contain obvious, high-signal spam vocabulary. This is expected and not very interesting: if a token is so strongly associated with spam that both attention and TF-IDF light up on it, it is likely a surface-level lexical cue.
- E_A (attention-only tokens) is where the interesting signal lives. These are tokens that attention prioritises but that the classical methods do not flag. On ham messages in particular, attention-exclusive tokens carry illocutionary weight—discourse markers, pragmatic particles, tokens involved in speech acts—that are invisible to bag-of-words methods because their importance depends on position and context, not raw frequency.
- E_R (robust-only tokens) captures the reverse: tokens with strong statistical class association that attention has learned to deprioritise, possibly because they are redundant given the sequential context the attention model has already encoded.

The core finding is an asymmetry: on spam, S is large and the methods mostly converge (the signal is lexically obvious). On ham, E_A grows and contains pragmatically meaningful tokens that only the attention models detect. This asymmetry is what makes the decomposition analytically useful rather than a bookkeeping exercise.

Caveats

None of the attribution methods used here are definitive:

- Attention weights are descriptive, not causal. A high-attention token may not be counterfactually important to the prediction.
- TF-IDF importance reflects corpus statistics, not per-message reasoning. A token can have a high LR coefficient globally but be irrelevant in a specific message context.
- Character n -grams operate below the word level and may flag orthographic artefacts rather than meaningful content.
- The top- k selection is a hard threshold; tokens just below the cutoff are discarded, which introduces sensitivity to the choice of k .

The cross-method comparison does not eliminate these limitations. What it does is ensure that findings which survive across methods are not artefacts of any single method’s blind spots. Convergence in S is more trustworthy than any individual attribution; divergence in E_A and E_R is a signal worth investigating precisely *because* the methods fail in different ways.

Set-theoretic reference

For completeness, the formal relationships:

Set difference is related to complement by $A \setminus R = A \cap R^c$, where R^c is the complement of R relative to U .

The **symmetric difference**

$$A \triangle R = E_A \cup E_R = (A \setminus R) \cup (R \setminus A)$$

captures total disagreement: the set of tokens that appear in exactly one of the two attribution families.

Spoken readout for presentation

- S or $A \cap R$: “the shared set” or “the overlap set”
- E_A or $A \setminus R$: “attention-only”
- E_R or $R \setminus A$: “robust-only”
- $A \triangle R$: “the total disagreement set”

For oral presentation: *shared tokens, attention-only tokens, and robust-only tokens*.

Compact paper-ready definition

No single token-attribution method provides ground-truth importance. Attention weights reflect where a model concentrates processing but are not reliably causal; TF-IDF-based importance captures corpus-level statistical association but is blind to word order and context; character n -gram features operate below the word level entirely. Because these methods fail in structurally different ways, cross-method comparison is more informative than any single method’s output alone.

Let U denote the set of tokens in a given message. Salient tokens are extracted by a top- k selection applied to each model’s token-importance scores. Let $A \subseteq U$ denote the union of top- k tokens selected by the attention-based models (LSTM, BiLSTM, and 2×BiLSTM with attention), and let $R \subseteq U$ denote the union of top- k tokens selected by the robust classical comparison pool (TF-IDF Naïve Bayes, TF-IDF Random Forest, TF-IDF Logistic Regression, and character n -gram TF-IDF Logistic Regression). We decompose message-level salience into the shared set $A \cap R$, the attention-exclusive set

$A \setminus R$, and the robust-exclusive set $R \setminus A$. These three regions are pairwise disjoint and partition the attributed region $A \cup R$. The symmetric difference $(A \setminus R) \cup (R \setminus A)$ captures total disagreement between the two attribution families.